

3(a)(i)	<i>accept from either column for mark and assume blood if not stated any three from</i> <table border="1"> <thead> <tr> <th data-bbox="338 284 1014 347">blood</th><th data-bbox="1014 284 1691 347">tissue fluid</th><th data-bbox="1691 284 1729 347"></th></tr> </thead> <tbody> <tr> <td data-bbox="338 347 1014 411">has red blood cells</td><td data-bbox="1014 347 1691 411">has no red blood cells</td><td data-bbox="1691 347 1729 411">;</td></tr> <tr> <td data-bbox="338 411 1014 579">more, white blood cells / leucocytes / named e.g. neutrophils, lymphocytes, monocytes or fewer macrophages</td><td data-bbox="1014 411 1691 579">fewer white blood cells / leucocytes / named e.g. neutrophils, lymphocytes, monocytes or more macrophages</td><td data-bbox="1691 411 1729 579">;</td></tr> <tr> <td data-bbox="338 579 1014 643">has platelets</td><td data-bbox="1014 579 1691 643">has no platelets</td><td data-bbox="1691 579 1729 643">;</td></tr> <tr> <td data-bbox="338 643 1014 882">has, more protein / large proteins or has, plasma proteins / named example e.g albumin / fibrinogen / globulin <i>these are large proteins</i></td><td data-bbox="1014 643 1691 882">has, fewer / no large, proteins or has no, plasma proteins / named example e.g. albumin / fibrinogen / globulin <i>only allow fewer plasma proteins if clear the proteins present are small enough to leave blood</i></td><td data-bbox="1691 643 1729 882">;</td></tr> <tr> <td data-bbox="338 882 1014 978">higher, concentration / AW, oxygen A more oxygenated</td><td data-bbox="1014 882 1691 978">lower, concentration / AW, oxygen A less oxygenated</td><td data-bbox="1691 882 1729 978">;</td></tr> <tr> <td data-bbox="338 978 1014 1074">higher, concentration / AW, glucose / amino acids / fatty acids</td><td data-bbox="1014 978 1691 1074">lower, concentration / AW, glucose / amino acids / fatty acids</td><td data-bbox="1691 978 1729 1074">;</td></tr> <tr> <td data-bbox="338 1074 1014 1209">lower, concentration / AW, carbon dioxide A urea, in liver / muscles I waste</td><td data-bbox="1014 1074 1691 1209">higher, concentration / AW, carbon dioxide A urea, in liver / muscles I waste</td><td data-bbox="1691 1074 1729 1209">;</td></tr> <tr> <td data-bbox="338 1209 1014 1281">higher pressure</td><td data-bbox="1014 1209 1691 1281">lower pressure</td><td data-bbox="1691 1209 1729 1281">;</td></tr> </tbody> </table>	blood	tissue fluid		has red blood cells	has no red blood cells	;	more, white blood cells / leucocytes / named e.g. neutrophils, lymphocytes, monocytes or fewer macrophages	fewer white blood cells / leucocytes / named e.g. neutrophils, lymphocytes, monocytes or more macrophages	;	has platelets	has no platelets	;	has, more protein / large proteins or has, plasma proteins / named example e.g albumin / fibrinogen / globulin <i>these are large proteins</i>	has, fewer / no large, proteins or has no, plasma proteins / named example e.g. albumin / fibrinogen / globulin <i>only allow fewer plasma proteins if clear the proteins present are small enough to leave blood</i>	;	higher, concentration / AW, oxygen A more oxygenated	lower, concentration / AW, oxygen A less oxygenated	;	higher, concentration / AW, glucose / amino acids / fatty acids	lower, concentration / AW, glucose / amino acids / fatty acids	;	lower, concentration / AW, carbon dioxide A urea, in liver / muscles I waste	higher, concentration / AW, carbon dioxide A urea, in liver / muscles I waste	;	higher pressure	lower pressure	;	3
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3(a)(ii)	lymph ; A lymphatic fluid	1																											

Question	Answer	Marks
3(b)	<i>any two from:</i> 1 can cross / AW, hydrophobic core (of bilayer) / phospholipid bilayer or can pass through gaps between, fatty acid tails / phospholipids ; 2 is, hydrophobic / lipid-soluble / non-polar / not ionic / not charged or not, hydrophilic / water-soluble / polar / ionic / charged ; 3 small size ;	2
3(c)	Reject if hormone S or receptor R described as an antigen or enzyme complementary (shape) to / only fits into / specificity for (binding to), R / receptor ;	1
3(d)(i)	nuclear pore ; A nuclear pores	1
3(d)(ii)	B = transcription ; C = translation ;	2
3(d)(iii)	(80S) ribosome ; I 70S	1

Question	Answer	Marks
3(e)	<i>any four from</i> 1 (formation of), secondary / 2° / second level, structure with, alpha-helix / α-helix / beta-pleated sheet(s) / β-pleated sheet(s) ; A β-conformation for β-pleated sheet I incorrect detail of bonds 2 folding / coiling, to form tertiary / 3° / third level, structure ; A <i>ref. to</i> 3D structure as alternative to, folding / coiling A secondary structure, folds / coils, to form tertiary structure R alpha helix / beta-pleated sheet, coils / folds 3 <i>two of</i> hydrogen bonds / ionic or electrostatic bonds / disulfide bonds or bridges / hydrophobic interactions ; I peptide bonds <i>as part of a list that contains two correct bond types</i> <i>these must be in context of tertiary structure</i> 4 (formation of), quaternary / 4° / fourth level structure ; A description e.g. polypeptides held together by, interactions / hydrogen bonds / ionic bonds / disulfide bonds R if peptide bonds included 5 (globular protein) has hydrophilic, amino acids / R-groups / side chains, to, outside / AW ; ora hydrophobic, amino acids / R-groups / side chains, towards centre / AW 6 example of other post-translational modification ; e.g. removal, methionine / met addition sugar group / glycosylation addition phosphate group / phosphorylation addition prosthetic group	4